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EXPERIMENT 18

Exp No:18 Experiment to understand KNN algorithm for a given data set

AIM:
To understand KNN algorithm for a given data set

DESCRIPTION:
Understand the KNN algorithm for the dataset given.

CODE:

```
import numpy as np

import pandas as pd

df = pd.read_csv('/content/Iris.csv')

df.info()

print(df.Species.value_counts())

print(df.head())

features = df.iloc[:, 1:-1].values

label = df.iloc[:, 5].values

from sklearn.model_selection import train_test_split

xtrain, xtest, ytrain, ytest = train_test_split( features, label, test_size=0.2,

random_state=42)

from sklearn.neighbors import KNeighborsClassifier

model_KNN = KNeighborsClassifier(n_neighbors=5)

model_KNN.fit(xtrain, ytrain)

print(model_KNN.score(xtrain, ytrain))
```

```
print(model_KNN.score(xtest, ytest))

from sklearn.metrics import confusion_matrix

print(confusion_matrix(ytest, model_KNN.predict(xtest)))

from sklearn.metrics import classification_report

print(classification_report(ytest, model_KNN.predict(xtest)))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
```

```
0.9666666666666667
1.0
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

RESULT:

Thus, the K-Nearest Neighbors (KNN) algorithm was implemented successfully to classify Iris flower species. The model achieved 96.67% accuracy on the test dataset and was evaluated using a confusion matrix and classification report.